Find the given term in the binomial expansion.

7. $(x+3)^5$ Find the 4th term.

8. $(3x-2)^6$ Find the 5th term.

Answers to 1.11A CA #1

1. a. $2x^2+8x+8$ b. 2 c. 8	2. a. $\frac{t^2-4t-5}{3t^3+3}$ b. $y=0$ c. $-\frac{5}{3}$	3. a. $2x(x+3)(x-3)$ b. $x=0,-3,3$ c. $[-3,0]\cup[3,\infty)$	4. a. $\frac{2(x-3)}{(x+3)(x-3)}$ b. none ($x=3$ is a hole) c. $x=-3$
5. $x^4-20x^3+150x^2-500x+625$		6. $32x^5+80x^4+80x^3+40x^2+10x+1$	
7. $270x^2$		8. $2160x^2$	

AP Precalc

Write your questions and thoughts here!

1.11B Polynomial Long Division & Slant Asymptotes

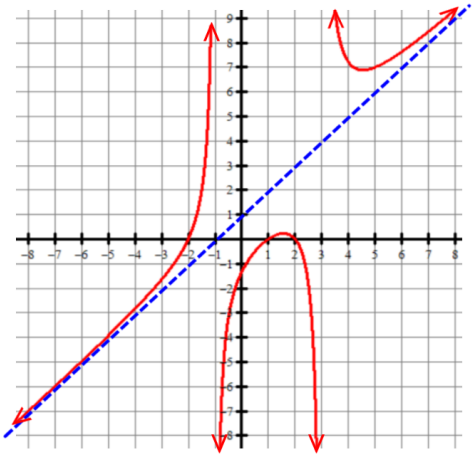
1.11B Notes

Slant Asymptotes

Degree of the numerator is one higher than degree of denominator!

Example #1

$$f(x) = \frac{x^3 - x^2 - 4x + 4}{x^2 - 2x - 3}$$



Example #2

$$f(x) = \frac{x^2 - x + 2}{x - 3}$$

Long Division

Synthetic Division

Write your questions and thoughts here!

Write the equation of the slant asymptote.

Example 3:

$$f(x) = \frac{4x^2 + 2x - 7}{2x + 3}$$

Example 4:

$$g(x) = \frac{6x^3 + 13x^2 - 5}{2x^2 + 3x}$$

End Behavior

$$g(x) = \frac{6x^3 + 13x^2 - 5}{2x^2 + 3x}$$

1.11B Polynomial Long Division and Slant Asymptotes

AP Precalculus

1.11B Practice

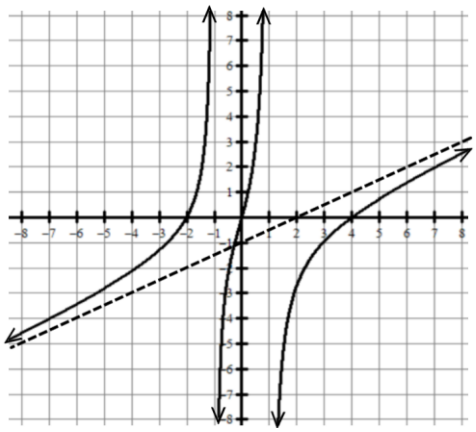
Divide the following using long division or synthetic division.

1. $\frac{3x^3-4x^2-3}{x^2+5x+1}$

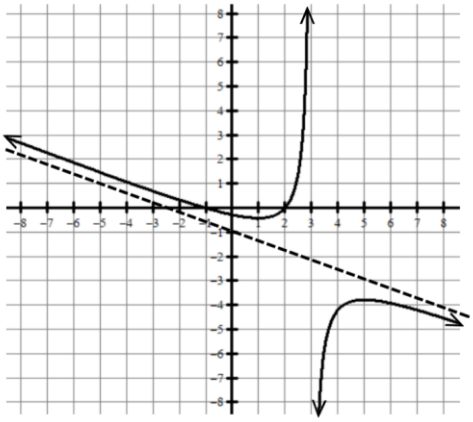
2. $\frac{x^3-4x^2+6x-4}{x-2}$

Use the graph of f to write the equation of the slant asymptote.

3.



4.



Determine if the following functions have a horizontal asymptote, slant asymptote, or neither.

5. $f(x)=\frac{4x^5-3x^3+4x+1}{5x^3-2x^2+1}$

Circle one:

The graph of f has a horizontal asymptote.

The graph of f has a slant asymptote.

The graph of f does not have a horziontal or slant asymptote.

6. $f(x)=\frac{2x^4+x^2+1}{3x^4-2x^2+5x}$

Circle one:

The graph of f has a horizontal asymptote.

The graph of f has a slant asymptote.

The graph of f does not have a horziontal or slant asymptote.

7. $f(x)=\frac{x^3+5x^2+x+2}{3x^4-2x^3+2x^2-3}$

Circle one:

The graph of f has a horizontal asymptote.

The graph of f has a slant asymptote.

The graph of f does not have a horziontal or slant asymptote.

Write the equation for the slant asymptote for the following functions.

8. $f(x) = \frac{x^3-2x^2-4x+1}{x^2-2x+1}$

9. $f(x) = \frac{x^2-9x+4}{x+6}$

10. $f(x) = \frac{4x^2+12x-6}{2x+1}$

11. $f(x) = \frac{9x^4-5x^2+3x-6}{3x^3-4x^2}$

Use the rational function to answer the following.

12.

$$f(x) = \frac{3x^3 - 12x}{x^2 - 2x - 8}$$

a. Domain:

b. Zero(s):

c. Hole(s):

d. Vertical Asymptote(s):

e. Horizontal Asymptote:

f. Slant Asymptote:

g. *y*-intercept:

h. *x*-intercept(s):

i. End Behavior:

Multiple Choice

13. The function f is a rational function. The quotient and remainder form of f is given by $f(x) = -2x + 1 + \frac{3x+4}{x^2-4x-12}$. Which describes the end behavior of f?

(A) $\lim_{x\to-\infty} f(x) = -\infty$ and $\lim_{x\to\infty} f(x) = -\infty$

(B) $\lim_{x\to-\infty} f(x) = \infty$ and $\lim_{x\to\infty} f(x) = \infty$

(C) $\lim_{x\to-\infty} f(x) = -\infty$ and $\lim_{x\to\infty} f(x) = \infty$

(D) $\lim_{x\to-\infty} f(x) = \infty$ and $\lim_{x\to\infty} f(x) = -\infty$

14. Which of the following is equivalent to $\frac{x^2+5x+2}{x+5}$?

(A) $x + 1$

(B) $x + 2$

(C) $x + \frac{2}{x+5}$

(D) $x + 1 - \frac{4}{x+5}$

15. The function f is given by $f(x) = \frac{6x^2+ax+2}{x+3}$ and has a slant asymptote of $y = 6x + 3$. What is the value of a?

(A) -4

(B) 12

(C) 15

(D) 21

1.11B Polynomial Long Division and Slant Asymptotes

AP Precalculus Name: _____ **CA #1**

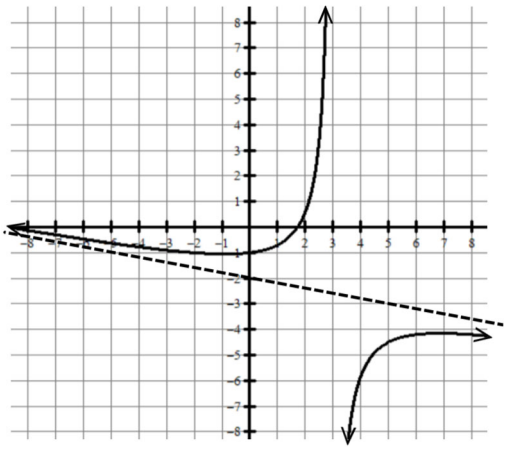
Divide the following using long division or synthetic division.

1. $\frac{2x^3-2x^2+10x-3}{x^2+3x-2}$

2. $\frac{x^4-4x^2-6x-4}{x-3}$

Use the graph of f to write the equation of the slant asymptote.

3.



Determine if the following functions have a horizontal asymptote, slant asymptote, or neither.

4. $f(x)=\frac{-3x^4+5x+1}{2x^2+1}$

Circle one:

The graph of f has a horizontal asymptote.

The graph of f has a slant asymptote.

The graph of f does not have a horziontal or slant asymptote.

5. $f(x)=\frac{2x^4+7x^2+1}{3x^5-5x^2+5x}$

Circle one:

The graph of f has a horizontal asymptote.

The graph of f has a slant asymptote.

The graph of f does not have a horziontal or slant asymptote.

6. $f(x)=\frac{x^3+4x^2+x+2}{2x^2+3x-3}$

Circle one:

The graph of f has a horizontal asymptote.

The graph of f has a slant asymptote.

The graph of f does not have a horziontal or slant asymptote.

Write the equation for the slant asymptote for the following functions.

7. $f(x) = \frac{9x^3-12x^2-5x+1}{3x^2-2x+1}$

8. $f(x) = \frac{2x^2+9x+6}{x+4}$

Answers to 1.11B CA #1

1. $2x - 8 + \frac{38x-19}{x^2+3x-2}$	2. $x^3 + 3x^2 + 5x + 9 + \frac{23}{x-3}$
3. $y = -\frac{1}{5}x - 2$	4. The graph of f does not have a horziontal or slant asymptote.
5. The graph of f has a horizontal asymptote.	6. The graph of f has a slant asymptote.
7. $y = 3x - 2$	8. $y = 2x + 1$

1.12A Translations of Functions

1.12A Notes

Additive Transformations

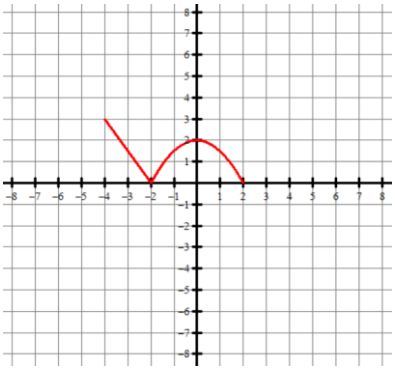
Translations =

Graphically

Example #1

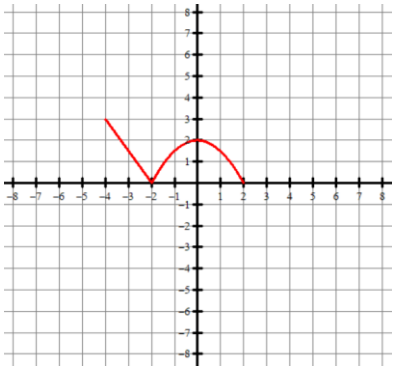
Given the graph f.

Let $g(x) = f(x) + 4$, graph $g(x)$



Given the graph f.

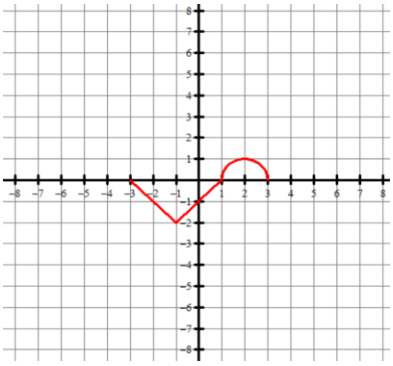
Let $g(x) = f(x + 4)$, graph $g(x)$



Example #2

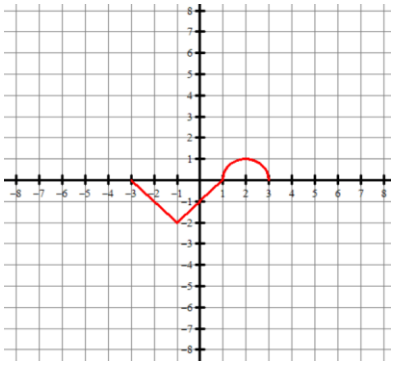
Given the graph f.

Let $g(x) = f(x) - 4$, graph $g(x)$.



Given the graph f.

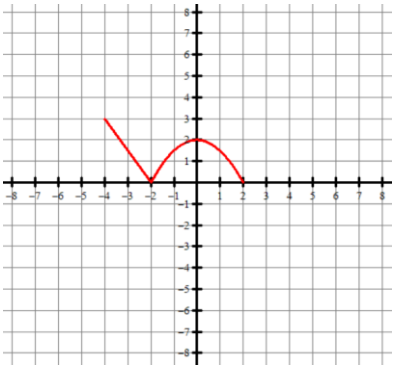
Let $g(x) = f(x - 4)$, graph $g(x)$.



Example #3 Vertical Reflection

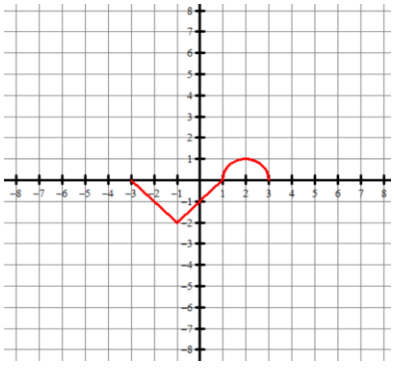
Given the graph f.

Let $g(x) = -f(x)$, graph $g(x)$.



Given the graph f.

Let $g(x) = -f(x - 2) + 5$, graph $g(x)$.



Write your questions and thoughts here!

Algebraically

Example 4:

Given $f(x) = x^2 - 3x + 2$

Let $g(x) = f(x) + 4$, find $g(x)$.

Given $f(x) = x^2 - 3x + 2$

Let $g(x) = f(x + 4)$, find $g(x)$.

Numerically

Example #5

Given the table of values for f.

x	$f(x)$
-2	21
-1	12
0	18
1	14
2	10

Let $g(x) = f(x) + 2$, find $g(2)$.

Given the table of values for f.

x	$f(x)$
0	-20
1	-12
2	0
3	8
4	14

Let $g(x) = f(x - 2) + 1$, find $g(4)$.

Domain and Range

Example #6

Given the graph for f has a domain of $[-4,3]$ and range of $(3, 9)$. Let $g(x) = -f(x + 5) + 2$. Find the domain and range of $g(x)$.

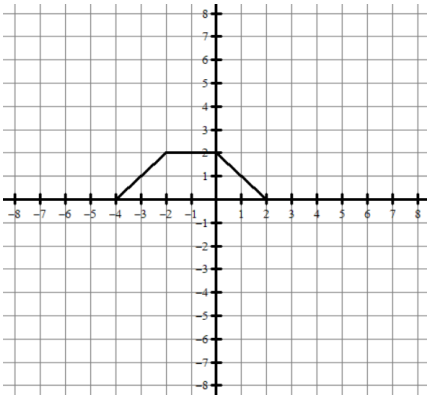
1.12A Translations of Functions

AP Precalculus

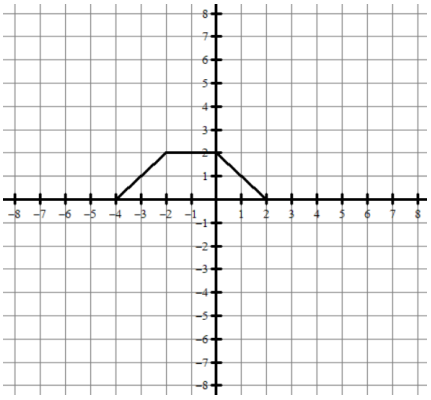
1.12A Practice

GRAPHICAL TRANSFORMATION. Use the graph of f to graph $g(x)$.

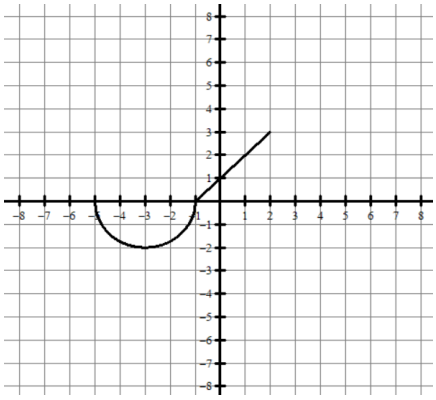
1. $g(x) = f(x-2) + 4$



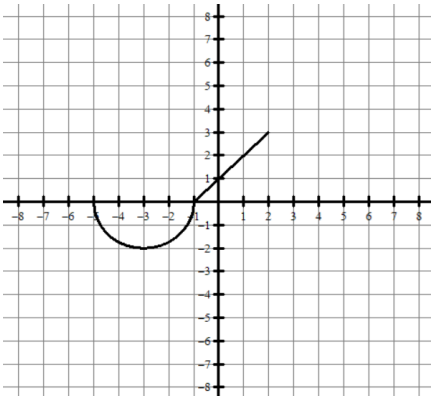
2. $g(x) = -f(x+3)$



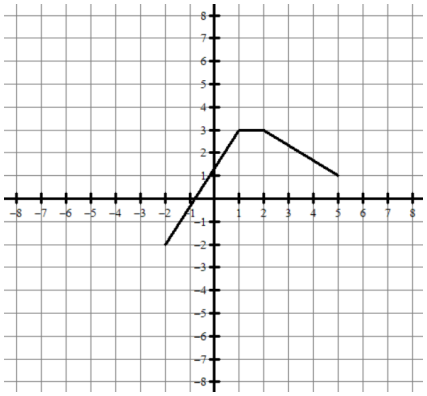
3. $g(x) = -f(x) + 5$



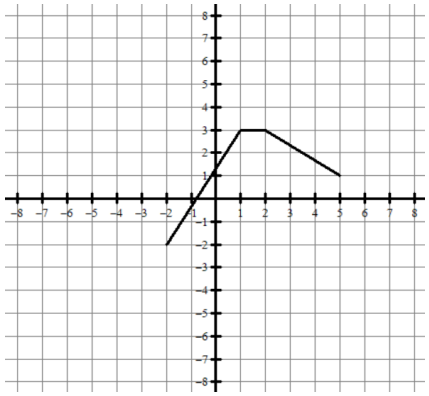
4. $g(x) = f(x-5) - 3$



5. $g(x) = f(x) - 4$



6. $g(x) = -f(x-3) + 1$



ALGEBRAIC TRANSFORMATION. Express the $g(x)$ in terms of x.

7. $f(x) = 4x + 3$

   $g(x) = f(x) + 5$, find $g(x)$.

8. $f(x) = 2x - 5$

   $g(x) = f(x+3) + 4$, find $g(x)$.

9. $f(x) = x^3 + 2x^2$

   $g(x) = -f(x) + 5$, find $g(x)$.

10. $f(x) = 2x^2 - 3x + 1$

    $g(x) = f(x-2) + 5$, find $g(x)$.

NUMERIC TRANSFORMATION. Use the table of values to answer the following.

11. Given the table of values for f.

x	$f(x)$
-6	2
-3	8
2	15
5	-2
8	-13

Let $g(x) = f(x) + 2$, find $g(5)$.

12. Given the table of values for f.

x	$f(x)$
0	0
1	2
2	4
3	8
4	16

Let $g(x) = f(x + 2) - 3$, find $g(1)$.

13. Given the table of values for f.

x	$f(x)$
-4	-32
-2	6
0	-8
2	21
4	14

Let $g(x) = -f(x - 2)$, find $g(4)$.

DOMAIN AND RANGE TRANSFORMATION. Find the domain and range of the transformed function.

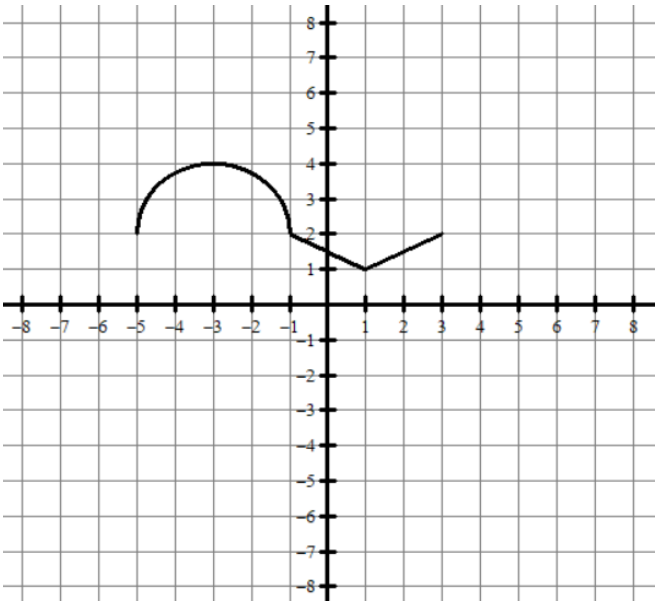
14.
Given the graph for f has a domain of $(-5,3]$ and range of $[-4,8]$.
Let $g(x) = f(x + 5)$.
Find the domain and range of $g(x)$.

15.
Given the graph for f has a domain of $(0,5)$ and range of $[-10,4]$.
Let $g(x) = f(x - 2) + 4$.
Find the domain and range of $g(x)$.

16.
Given the graph for f has a domain of $[-2,4]$ and range of $(-1,8)$.
Let $g(x) = -f(x + 3) + 5$.
Find the domain and range of $g(x)$.

Use the graph f to answer the following.

17.



Let the $g(x) = -f(x + 3) + 2$

a. Graph the $g(x)$.

b. State the domain of $g(x)$.

c. State the range of $g(x)$.

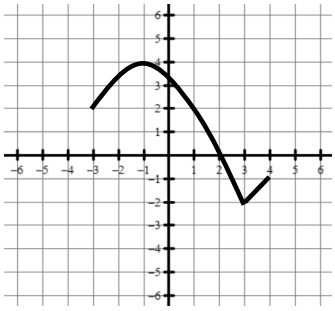
d. Find $g(-2)$.

e. Find the zeroes of $g(x)$.

f. Find the *y*-intercept of $g(x)$.

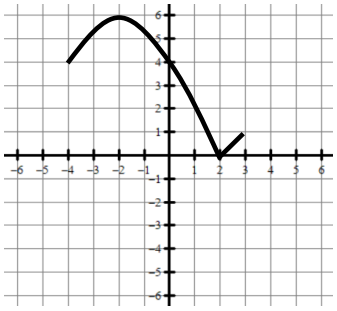
Multiple Choice

18. The graph of $y = f(x)$ is shown for $-3 \leq x \leq 4$.

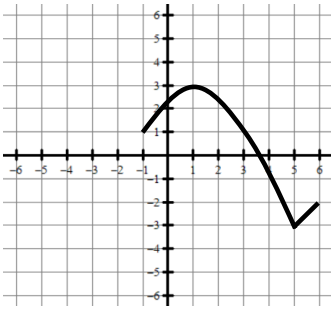


Which of the following is the transformed graph for $y = f(x + 2) - 1$?

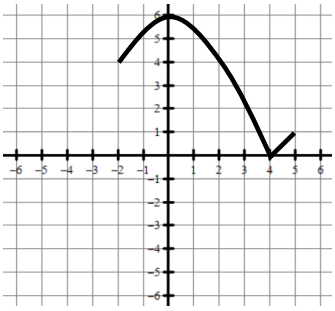
(A)



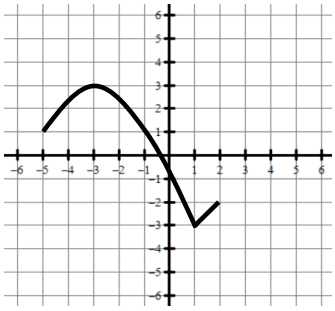
(B)



(C)



(D)



---

19. The functions f and g are defined for all real numbers such that $g(x) = -f(x) + 5$. Which of the following sequences of transformations maps the graph of f to the graph of g in the same xy-plane?

(A) A horizontal translation of the graph of f by 5 units, followed by a vertical reflection of the graph of f.

(B) A vertical translation of the graph of f by 5 units, followed by a vertical reflection of the graph of f.

(C) A vertical reflection of the graph of f, followed by a horizontal translation of the graph of f by 5 units.

(D) A vertical reflection of the graph of f, followed by a vertical translation of the graph of f by 5 units.

---

20. The function f is given by $f(x) = -x^2 + 3x + 2$. The graph of which of the following functions is the image of the graph of f after a vertical translation of the graph of f by 4 units ?

(A) $m(x) = -(x + 4)^2 + 3(x + 4) + 2$, because this is an additive transformation of f that results from adding to each input value of x.

(B) $n(x) = -(x - 4)^2 + 3(x - 4) + 2$, because this is an additive transformation of f that results from adding to each input value of x.

(C) $p(x) = -x^2 + 3x + 6$, because this is an additive transformation of f that results from adding to the $f(x)$.

(D) $q(x) = -x^2 + 3x - 2$, because this is an additive transformation of f that results from adding to the $f(x)$.

1.12A Translations of Functions

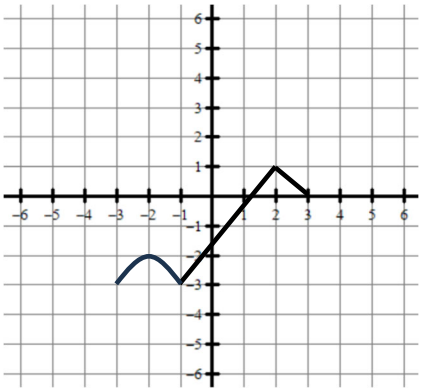
AP Precalculus

Name: _____

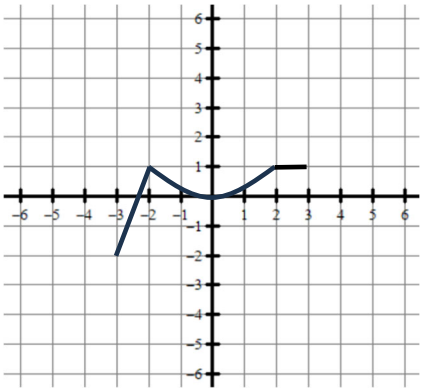
CA #1

GRAPHICAL TRANSFORMATION. Use the graph of f to graph $g(x)$.

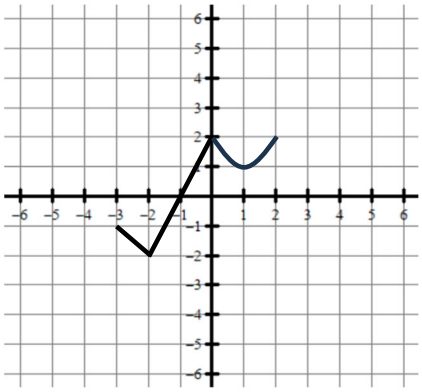
1. $g(x) = f(x-2)+4$



2. $g(x) = -f(x+3)$



3. $g(x) = f(x-3)-1$



ALGEBRAIC TRANSFORMATION. Express the $g(x)$ in terms of x.

4. $f(x) = -x^2+3$

$g(x) = f(x)+2$, find $g(x)$.

5. $f(x) = 2x^2+3$

$g(x) = f(x+2)$, find $g(x)$.

6. $f(x) = 2x-5$

$g(x) = -f(x+1)+4$, find $g(x)$.

NUMERIC TRANSFORMATION. Use the table of values to answer the following.

7. Given the table of values for f.

x	$f(x)$
-2	12
-1	18
0	5
1	-12
2	-3

Let $g(x) = f(x)+2$, find $g(-1)$.

8. Given the table of values for f.

x	$f(x)$
0	12
1	9
2	6
3	3
4	0

Let $g(x) = f(x+2)-3$, find $g(2)$.

9. Given the table of values for f.

x	$f(x)$
-4	2
-2	-6
0	-12
2	2
4	14

Let $g(x) = -f(x-2)+1$, find $g(0)$.

DOMAIN AND RANGE TRANSFORMATION. Find the domain and range of the transformed function.

10. Given the graph for f has a domain of $(-1,3)$ and range of $[5,10]$.

Let $g(x) = f(x+3)$.

Find the domain and range of $g(x)$.

11. Given the graph for f has a domain of $(-10,4]$ and range of $[-3,6]$.

Let $g(x) = f(x+1) + 4$.

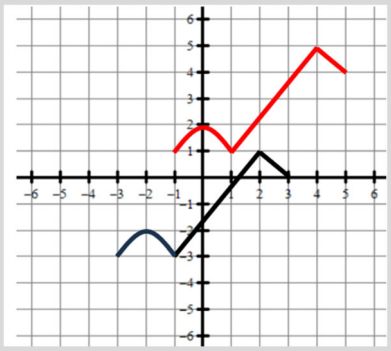
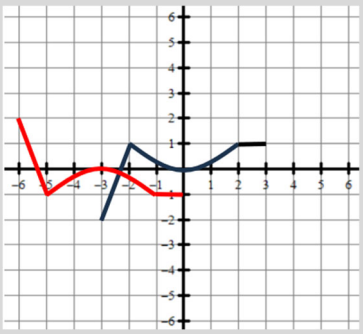
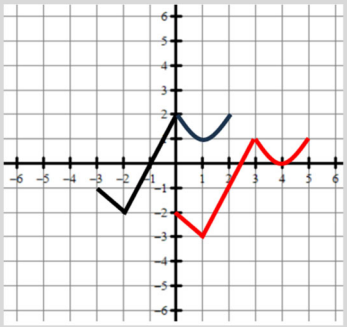
Find the domain and range of $g(x)$.

12. Given the graph for f has a domain of $[-2,5]$ and range of $(-2,8)$.

Let $g(x) = -f(x+3) - 6$.

Find the domain and range of $g(x)$.

Answers to 1.12A CA #1

1. 	2. 	3. 
4. $g(x) = -x^2 + 5$	5. $g(x) = 2x^2 + 8x + 11$	6. $g(x) = -2x + 7$
7. 20	8. -3	9. 7
10. Domain: $(-4,0)$ Range: $[5,10]$	11. Domain: $(-11,3]$ Range: $[1,10]$	12. Domain: $[-5,2]$ Range: $(-14,-4)$

1.12B Dilations of Functions

1.12B Notes

Multiplicative Transformations

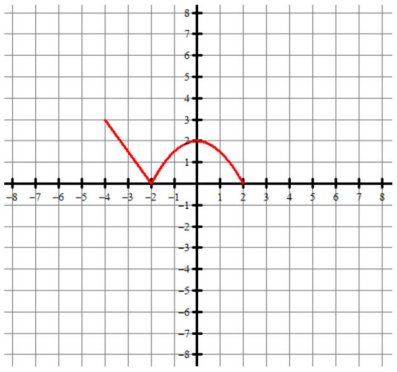
Dilations =

Graphically

Example #1 Vertical Dilations

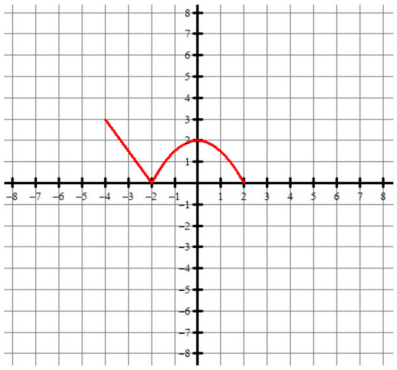
Given the graph f.

Let $g(x) = 2f(x)$, graph $g(x)$



Given the graph f.

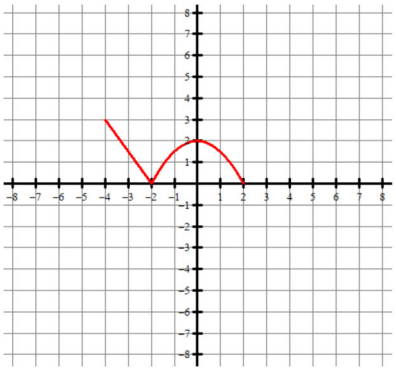
Let $g(x) = \frac{1}{2}f(x)$, graph $g(x)$



Example #2 Horizontal Dilations

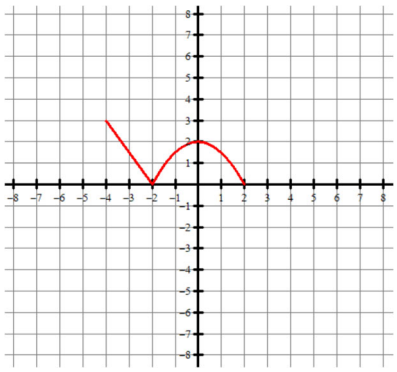
Given the graph f.

Let $g(x) = f(2x)$, graph $g(x)$.



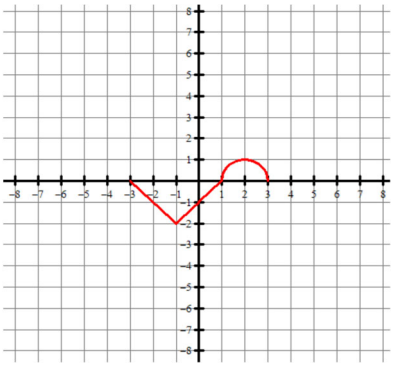
Given the graph f.

Let $g(x) = f\left(\frac{1}{2}x\right)$, graph $g(x)$.



Given the graph f.

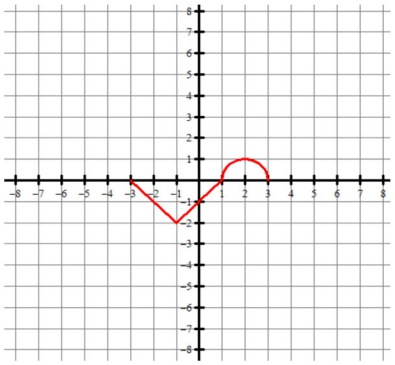
Let $g(x) = f(-x)$, graph $g(x)$.



Example #3

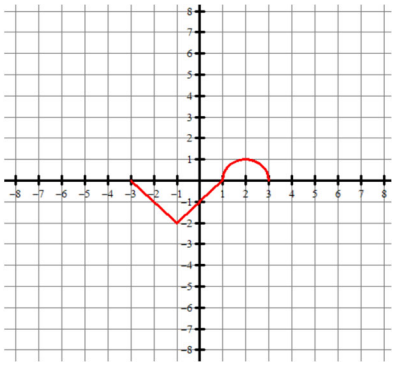
Given the graph f.

Let $g(x) = 3f(x) + 2$, graph $g(x)$.



Given the graph f.

Let $g(x) = -f\big(2(x+3)\big) + 1$, graph $g(x)$.



Write your questions and thoughts here!

Algebraically

Example 4:

Given $f(x) = x^2 - 3x + 2$

Let $g(x) = 2f(x) + 4$, find $g(x)$.

Given $f(x) = 6x + 3$

Let $g(x) = \frac{1}{3}f(x + 2)$, find $g(x)$.

Numerically

Example #5

Given the table of values for f.

x	$f(x)$
-2	21
-1	12
0	18
1	14
2	10

Let $g(x) = 5f(x) - 1$, find $g(-1)$.

Given the table of values for f.

x	$f(x)$
0	-20
1	-12
2	0
3	8
4	14

Let $g(x) = f(3(x - 1)) + 2$, find $g(2)$.

Domain and Range

Example #6

Given the graph for f has a domain of $[-2,5]$ and range of $(4, 9)$. Let $g(x) = 3f(x) + 2$. Find the domain and range of $g(x)$.

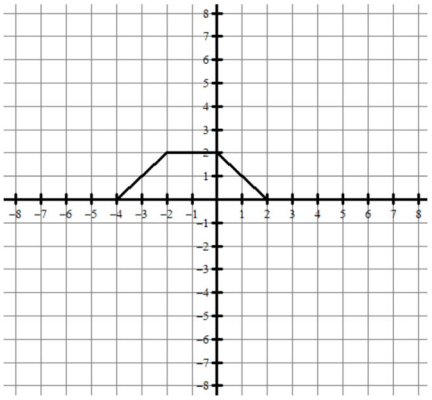
1.12B Dilations of Functions

AP Precalculus

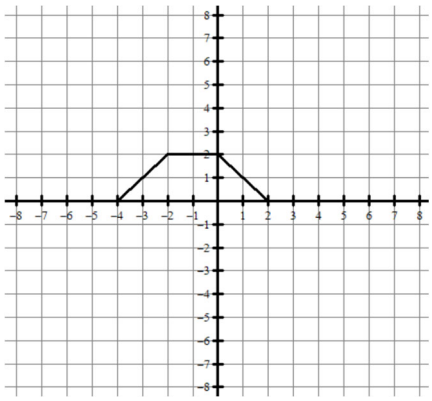
1.12B Practice

GRAPHICAL TRANSFORMATION. Use the graph of f to graph $g(x)$.

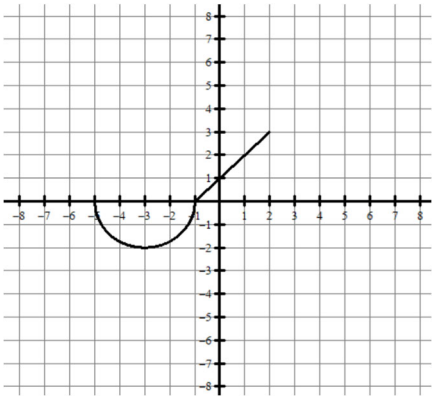
1. $g(x) = 3f(x) - 5$



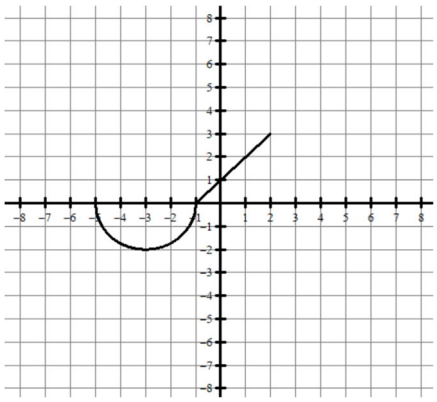
2. $g(x) = f\left(\frac{1}{2}x\right)$



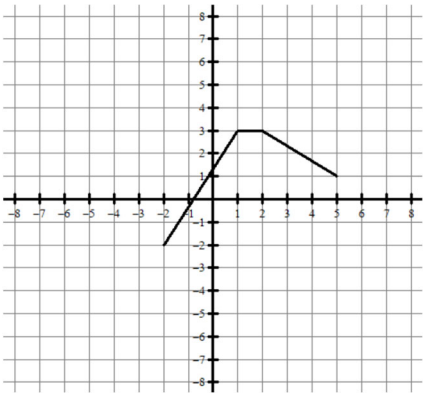
3. $g(x) = -f(x-2)$



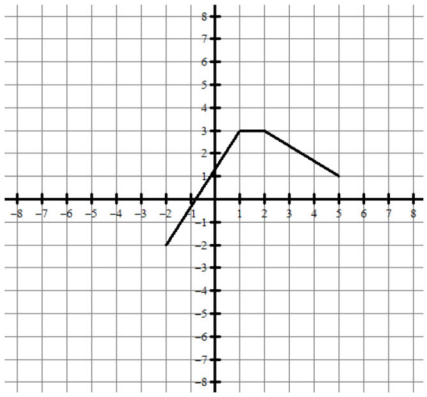
4. $g(x) = \frac{1}{2}f(x)$



5. $g(x) = f(-x) - 4$



6. $g(x) = f(2(x+1))$



ALGEBRAIC TRANSFORMATION. Express the $g(x)$ in terms of x.

7. $f(x) = 4x + 3$

   $g(x) = 3f(x) + 5$, find $g(x)$.

8. $f(x) = 2x + 6$

   $g(x) = f\left(\frac{1}{2}(x+2)\right) - 1$, find $g(x)$.

9. $f(x) = 4x - 5$

   $g(x) = -2f(x+1) + 5$, find $g(x)$.

10. $f(x) = 2x^2 - 3x + 1$

    $g(x) = f(2x) + 3$, find $g(x)$.

NUMERIC TRANSFORMATION. Use the table of values to answer the following.

11. Given the table of values for f.

x	$f(x)$
-9	2
-3	8
2	15
6	-2
8	-13

Let $g(x) = f(3x) + 1$, find $g(2)$.

12. Given the table of values for f.

x	$f(x)$
0	0
1	2
2	4
3	8
4	16

Let $g(x) = 2f(x + 2) - 3$, find $g(1)$.

13. Given the table of values for f.

x	$f(x)$
-8	-32
-2	6
0	-8
2	21
8	14

Let $g(x) = 4f(-x)$, find $g(2)$.

DOMAIN AND RANGE TRANSFORMATION. Find the domain and range of the transformed function.

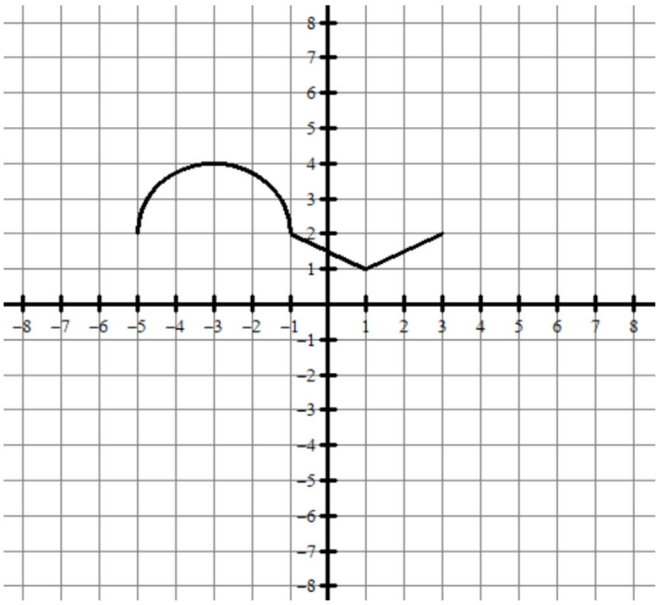
14.
Given the graph for f has a domain of $(-6,8]$ and range of $[-4,8]$.
Let $g(x) = -3f(2x)$.
Find the domain and range of $g(x)$.

15.
Given the graph for f has a domain of $(0,5)$ and range of $[-10,4]$.
Let $g(x) = 2f(x - 3) + 4$.
Find the domain and range of $g(x)$.

16.
Given the graph for f has a domain of $[-2,4]$ and range of $(-6,8)$.
Let $g(x) = f\left(\frac{1}{2}x\right) + 5$.
Find the domain and range of $g(x)$.

Use the graph f to answer the following.

17.



Let the $g(x) = 2f(x - 3) - 1$

a. Graph the $g(x)$.

b. State the domain of $g(x)$.

c. State the range of $g(x)$.

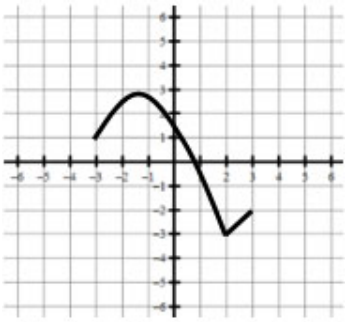
d. Find $g(-2)$.

e. Find the zeroes of $g(x)$.

f. Find the y-intercept of $g(x)$.

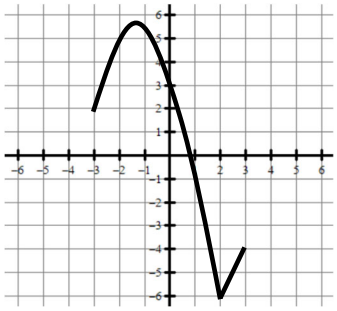
Multiple Choice

18. The graph of $y = f(x)$ is shown for $-3 \leq x \leq 4$.

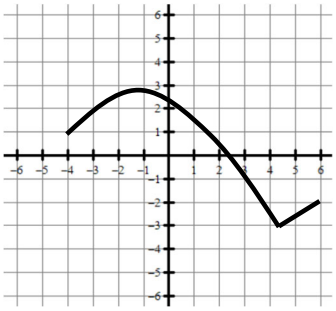


Which of the following is the transformed graph for $y = f(2x)$?

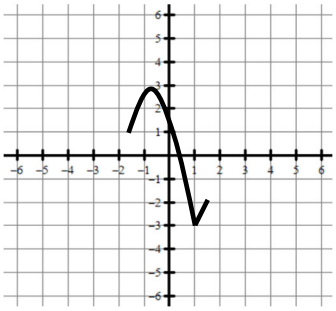
(A)



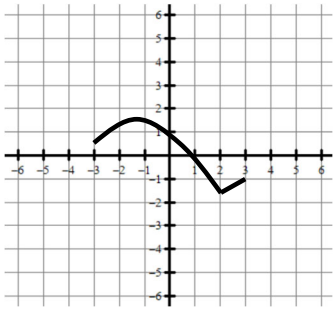
(B)



(C)



(D)



19. The table gives values for a polynomial function f at selected values of x.

x	-2	0	2	4	8	16
$f(x)$	38	12	-24	-32	-12	-48

In the *xy*-plane, the graph of g is constructed by applying three transformations to the graph of f in this order: a horizontal dilation of $\frac{1}{2}$, a vertical dilation of 4 and vertical translation by 6 units. What is the value of $g(4)$?

(A) 3

(B) -42

(C) -90

(D) 0

20. The function g is defined by f such that $g(x) = 3f(2x) - b$. If $f(2) = a, f(4) = 8, g(1) = 7$ and $g(2) = 19$ what is the value of a ?

(A) 6

(B) 4

(C) 2

(D) -2

1.12B Dilations of Functions

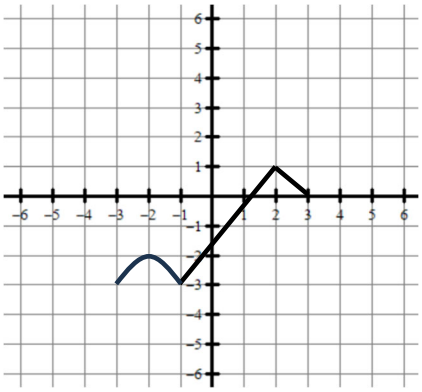
AP Precalculus

Name: _____

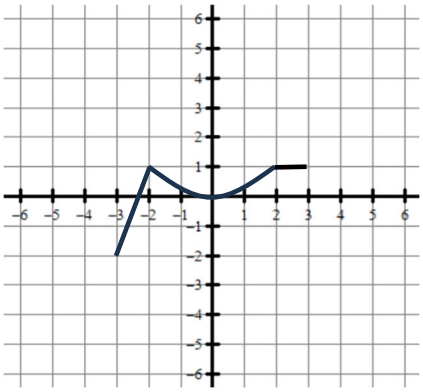
CA #1

GRAPHICAL TRANSFORMATION. Use the graph of f to graph $g(x)$.

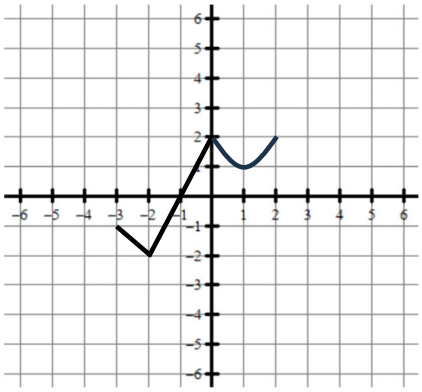
1. $g(x) = 2f(-x) + 3$



2. $g(x) = f\left(\frac{1}{2}x\right) - 1$



3. $g(x) = f(2(x-4))$



ALGEBRAIC TRANSFORMATION. Express the $g(x)$ in terms of x.

4. $f(x) = -x^2 + 3$

$g(x) = 3f(x) + 2$, find $g(x)$.

5. $f(x) = 2x^2 + 3$

$g(x) = 2f(3x)$, find $g(x)$.

6. $f(x) = 2x - 5$

$g(x) = -f\left(\frac{1}{2}x\right) + 4$, find $g(x)$.

NUMERIC TRANSFORMATION. Use the table of values to answer the following.

7. Given the table of values for f.

x	$f(x)$
-2	12
-1	18
0	5
1	-12
2	-3

Let $g(x) = f(2x) + 2$,

find $g(-1)$.

8. Given the table of values for f.

x	$f(x)$
0	12
1	9
2	6
3	3
4	0

Let $g(x) = 4f(x+2) - 3$,

find $g(2)$.

9. Given the table of values for f.

x	$f(x)$
-2	2
-1	-6
0	-12
1	2
2	14

Let $g(x) = -2f\left(\frac{1}{2}(x-2)\right) + 1$,

find $g(0)$.

DOMAIN AND RANGE TRANSFORMATION. Find the domain and range of the transformed function.

10. Given the graph for f has a domain of $(-1,3)$ and range of $[5,10]$.

Let $g(x) = 2f(x+3)$.

Find the domain and range of $g(x)$.

11. Given the graph for f has a domain of $(-10,4]$ and range of $[-3,6]$.

Let $g(x) = -f(2x)$.

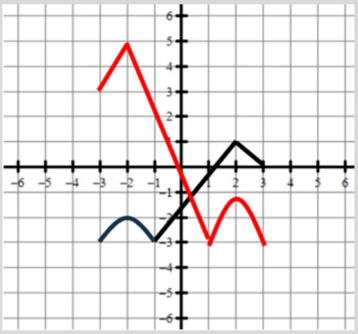
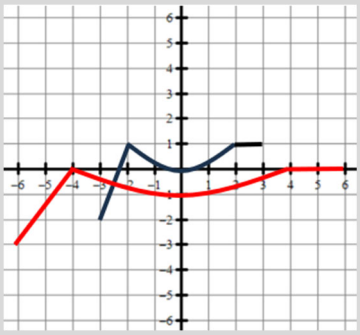
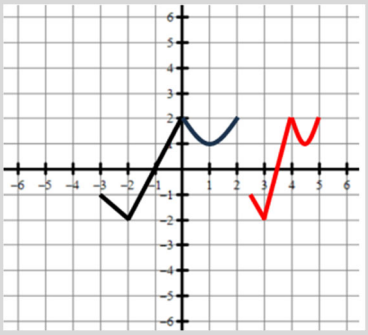
Find the domain and range of $g(x)$.

12. Given the graph for f has a domain of $[-2,5]$ and range of $(-2,8)$.

Let $g(x) = \frac{1}{2}f(x) - 6$.

Find the domain and range of $g(x)$.

Answers to 1.12B CA #1

1. 	2. 	3. 
4. $g(x) = -3x^2 + 11$	5. $g(x) = 36x^2 + 6$	6. $g(x) = -x + 9$
7. 14	8. -3	9. 13
10. Domain: $(-4, 0)$ Range: $[10, 20]$	11. Domain: $(-5, 2]$ Range: $[-6, 3]$	12. Domain: $[-2, 5]$ Range: $(-7, -2)$

AP Precalc

Write your questions and thoughts here!

1.13 Function Model Selection

1.13 Notes

Selecting Models

Look at context, data, graphs and select the appropriate function to model the situation.
Models may have restricted domain and range!!!

Bean

day	1	2	3	4	5	6
stamps	15	20	25	30	35	40

Kelly

day	1	2	3	4	5	6
stamps	6	8	12	18	26	36

Sully

day	1	2	3	4	5	6
stamps	8	13	30	65	124	213

Brust

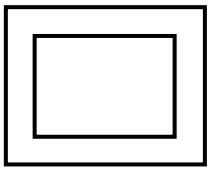
day	1	3	4	5	7	8
stamps	83	89	92	95	101	104

Geometry

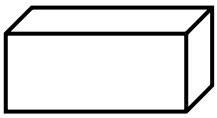
Perimeter



Area



Volume



Write your questions and thoughts here!

Restricted Domain and Range

A bottle rocket is set up on a picnic table. When the firework detonates it will shoot straight up and come back down. The height of the bottle rocket is modeled by $h(t) = -16t^2 + 64t + 3$ where *t* stands for time in seconds and *h* is height of the rocket in feet.

a. Find $h(2)$. What does it mean in this context?

b. What is the restricted domain?

c. What is the restricted range?

d. What is the average rate of change for the interval [2,3] ?

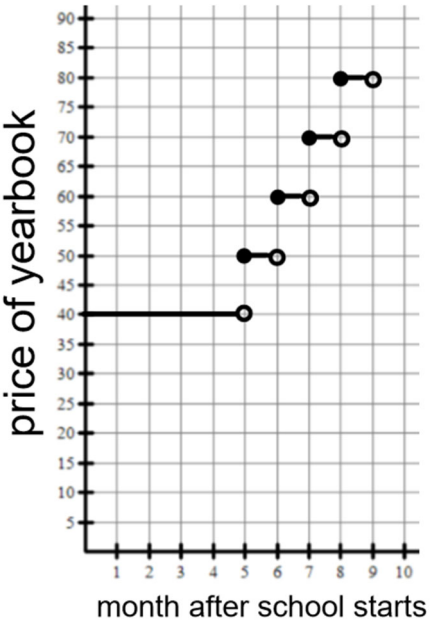
Piecewise Functions

The graph of f below shows the price of Generic High School yearbook. The yearbook sells for $40 the first 5 months of school. After that, the price increases every month.

a. What is the domain in this context?

b. What is the range in this context?

c. Find $f(5.5)$. What does it mean in this context?



1.13 Function Model Selection

AP Precalculus

1.13 Practice

Select the appropriate model for the data (linear/quadratic/cubic). Explain why it models the data.

1.

x	1	2	3	4	5	6
y	4	15	50	121	240	419

2.

x	1	2	3	4	5	6
y	49	46	43	40	37	34

3.

x	1	2	3	4	5	6
y	119	110	95	74	47	14

4.

x	2	4	5	6	7	8
y	23	81	122	171	228	293

CALCULATOR ACTIVE. Use the model to answer the questions in context.

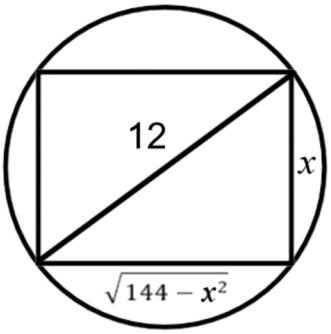
5. A patient receives a dose of painkiller. The function $p(t) = \frac{2t^2+10t}{t^3+1}$ models the amount of painkiller in the blood stream over time, where t is time in hours and p is painkiller in milligrams.

   a. Find $p(2)$. Explain your solution in context.

   b. What is the average rate of change from $t = 1$ to $t = 2$? Explain your solution in context.

   c. What is the maximum amount of painkiller in the patient's bloodstream?

6. A rectangle is inscribed in a circle with diameter of 12 cm. The width of the rectangle is x cm. The function $A(x) = x\sqrt{144 - x^2}$ models the area of the rectangle.



   a. What is the restricted domain of the function?

   b. What is the restricted range of the function?

   c. What is the maximum area of the rectangle?

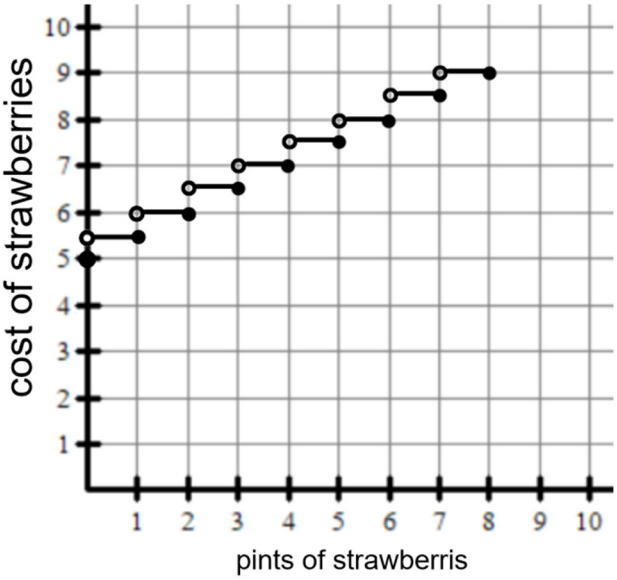
Use the graph of the piecewise function to answer the questions in context.

7. Generic Strawberry Fields allows customers to pay \$5 to pick strawberries plus 50 cents for every pint or partial pint of strawberries that they pick. There is a limit of 8 pints per customer. The piecewise function f shown below models the price of strawberries picked.

a. What is the domain in this context?

b. The range can be represented by $y = 5 + 0.5x$ where x is all integers in the domain. Explain why.

c. Find $f(3.5)$. What does it mean in this context?



---

8. The piecewise function f shown below models the insulin levels of a patient over time where $x = 0$ represents 8:00.

a. What is the domain in this context?

b. What is the range in this context?

c. Find $f(3)$. What does it mean in this context?

d. Find average rate of change from [0, 3]. Explain the meaning in this context.



Multiple Choice - CALCULATOR ACTIVE

For questions 9 and 10, use the table of values for polynomial f at selected values of x.

x	2	4	6	8	10	12
$f(x)$	97	405	977	1861	3105	4757

9. Which of the following statements are true ?

(A) The function is quadratic because the second differences are a nonzero constant.

(B) The function is quadratic because the third differences are a nonzero constant.

(C) The function is cubic because the second differences are a nonzero constant.

(D) The function is cubic because the third differences are a nonzero constant.

10. The average rate of change from $x = -22$ to $x = 4$ is 34. Which statement best describes the $f(x)$?

(A) There must be at least one zero on the interval $[-22, 4]$ because $f(-22)$ is negative.

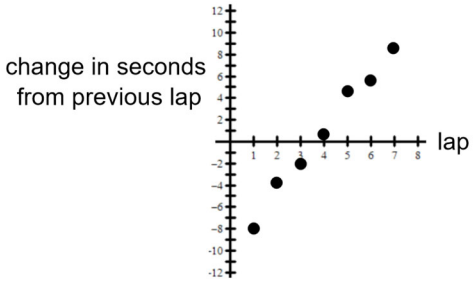
(B) The number of zeros on the interval $[-22, 4]$ cannot be determined because $f(-22)$ is positive.

(C) There must be at least one zero on the interval $[-22, 4]$ because the average rate of change is negative.

(D) There is at least one zero in that interval$[-22, 4]$ because the average rate of change is positive.

---

11. A track athlete is running laps. The graph shows the average rate of change from the previous lap.



A function model T is constructed for the time of each lap. Which of the following statements best supports the selection of the model of a model for T

(A) Since the rate of change is roughly linear, a linear model is best for T.

(B) Since the rate of change is roughly linear, a quadratic model is best for T.

(C) Since the rate of change is roughly linear, a cubic model is best for T.

(D) Since the rate of change is negative and positive, a quadratic model is best for T.

1.13 Function Model Selection

AP Precalculus Name: _____ **CA #1**

Select the appropriate model for the data (linear/quadratic/cubic). Explain why it models the data.

1.

x	1	2	3	4	5	6
y	1	-4	-15	-32	-55	-84

2.

x	2	4	6	8	10	12
y	-13	-8	-3	2	7	12

3.

x	0	1	2	3	4	5
y	5	3	11	35	81	155

4.

x	2	4	5	6	7	8
y	5	-7	-16	-27	-40	-55

CALCULATOR ACTIVE. Use the model to answer the questions in context.

5. A diver jumps from a cliff into the water below. The function $h(t) = -16t^2 + 12t + 120$ models the height of the diver over time, where *t* is time in seconds and *h* is the height in feet.

   a. Find $h(2)$. Explain your solution in context.

   b. What is the average rate of change from $t = 1$ to $t = 2$? Explain your solution in context.

   c. What is the maximum height of the diver?

   d. What is the restricted domain in this context?

   e. What is the restricted range in this context?

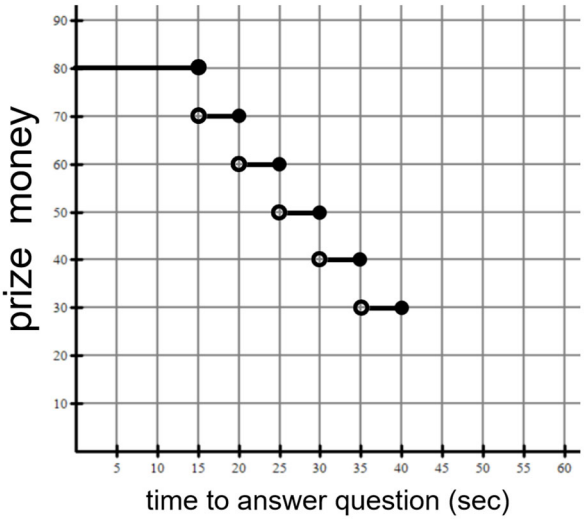
Use the graph of the piecewise function to answer the questions in context.

6. A radio station is having a contest. The money won by listeners depends on how long it takes them to correctly answer the question.

   a. What is the domain in this context?

   b. What is the range in this context?

   c. Find $f(30)$. What does it mean in this context?



Answers to 1.13 CA #1

1. Quadratic, second difference of -6	2. Linear, first difference of $\frac{5}{2}$
3. Cubic, third difference of 6	4. Quadratic, second difference of -8
5. a. $h(2) = 80$ In 2 seconds the height of the diver is 80 ft b. -36 feet per second c. 122.25 feet d. Domain: $[0, 3.139]$ e. Range: $[0, 122.25]$	6. a. Domain: $[0, 40]$ b. Ramge: $\{80, 70, 60, 50, 40, 30\}$ c. $f(30) = 50$ If it takes the listener 30 seconds to answer the question, they get \$50

AP Precalc

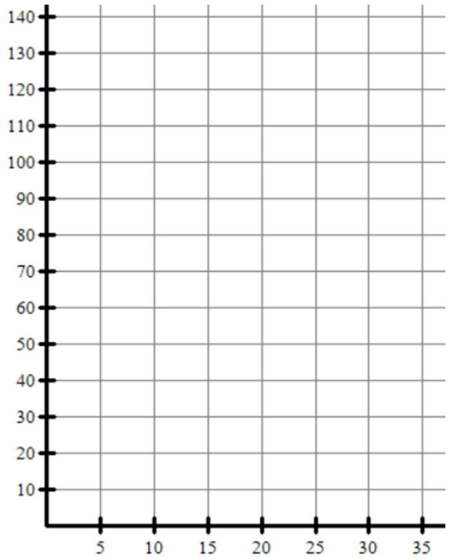
Write your questions and thoughts here!

1.14 Function Model Construction

1.14 Notes

Jeff Bezos Net Worth

Year	Net worth (billions)
1994	0
1998	12
2002	10
2008	2
2015	50
2018	100
2020	136



Use regression to write an equation for the best fit curve.

Use your model to find the average rate of change from 2008 to 2022.

Types of Regression

Linear	Quadratic	Cubic	Quartic
Exponential	Logarithmic	Logistic	Sine

Write your questions and thoughts here!

Inversely Proportional

The force in newtons exerted by a magnet on a metal object is inversely proportional to the square of the distance in centimeters. If a magnet exerts 60 newtons on a metal object 2 cm away, what will be the distance when the force is 10 newtons?

Piecewise Functions Algebraically

The following model tracks the amount of accumulating snow during a snowstorm where *x* is the time in hours and *f* is depth of the snow in inches.

$$f(x) = \begin{cases} \frac{1}{4}x^2 + 2x\,, & 0 \le x \le 4 \\ 12\,, & 4 < x \le 6 \\ x + 3\,, & 6 < x \le 9 \end{cases}$$

a. Find the $f(7)$. Explain what it means in this context.

b. Find the average rate of change from 4 to 5 hours.

1.14 Function Model Construction

AP Precalculus

1.14 Practice

CALCULATOR ACTIVE Perform the indicated regression and answer the questions.

1. Run a Quadratic Regression.

x	1	2	3	4	5	6
$f(x)$	10	6	5	5.5	6	11

a. Write the equation of the regression curve.

b. Use your equation to predict $f(2.5)$.

---

2. Run a Quartic Regression.

x	-6	-4	-2	2	4	6
$g(x)$	2700	928	80	66	500	2450

a. Write the equation of the regression curve.

b. Use your equation find average rate of change from $x = 3$ to $x = 6$.

CALCULATOR ACTIVE Graph the data and choose the regression that best fits the data.

3. The data shows the salary in thousands of dollars for employees given their years experience.

Experience (years)	Salary (thousands)
2	48
4	54
5	58
8	68
9	73
12	80
15	88

a. Is the data Linear, Quadratic, or Cubic ?

b. Write the equation of the regression curve.

c. Use your equation to predict the salary of an employee with 11 years experience.

---

4. The data shows poll ratings of a politician over time.

month	1	2	3	4	5	6	7	8	9	10	11	12
rating	12	18	25	32	41	37	30	27	35	42	49	56

a. Is the data Linear, Quadratic, or Cubic ?

b. Write the equation of the regression curve.

c. Use your equation to find the average rate of change from May to October.

CALCULATOR ACTIVE Construct an inversely proportional model to answer the following.

5. The time taken for a water heater to boil water is inversely proportional to the power used in watts. It takes 2000 watts to boil water in 240 seconds. Find the time it takes when using 900 watts.

6. The data is inversely proportional. Find the value of a.

x	3	10	12
$f(x)$	16	4.8	a

7. Price is inversely proportional to time squared. If on day 2 the price is \$4, what is the price on day 4?

8. The cost per cookie for making cookies is inversely proportional to the square root of the number of cookies made. If it costs \$2 each to make 9 cookies, how much would it cost for each cookie to make 25 cookies?

CALCULATOR ACTIVE Use the piecewise function to answer the following.

9.

$$f(x) = \begin{cases} 9x - x^2, & -2 \le x \le 3 \\ 2x - 17, & 3 < x < 10 \\ 5, & 10 \le x \le 12 \end{cases}$$

a. Find $f(1)$.

b. Find $f(10)$.

10. A rental car costs \$20 to rent plus 70 cents every mile driven for the first 100 miles. Miles driven over 100 only cost 40 cents.

$$C(m) = \begin{cases} 20 + 0.7m, & 0 \le m \le 100 \\ 0.4(m - 100) + 90, & m > 100 \end{cases}$$

a. Find $C(128)$.

b. How many miles did a customer drive that spent 24 dollars on their rental car?

Multiple Choice – Calculator Active

11. The table below shows the average price of a movie ticket during certain years.

year	2012	2015	2018	2019	2020	2022
price	7.96	8.17	8.97	9.11	9.16	10.12

A linear regression is used to construct a function model P that models the price over the given years. If $t = 1$ corresponds to 2012, $t = 4$ corresponds to 2015, and this pattern continues, which of the following defines function P?

(A) $P(t) = 0.206x + 7.541$

(B) $P(t) = 0.397x + 7.524$

(C) $P(t) = 0.206x + 7.747$

(D) $P(t) = 0.206x - 407.07$

---

12. The weight of an object is inversely proportional to the square of the distance between an object and the center of the earth. This relationship is modeled by the function W, where $W(d) = \frac{2.944 \times 10^9}{d^2}$ for distance, d, measured in feet, and weight where $W(d)$ measured in pounds. What is the average rate of change, in pounds per foot, if the distance between an object and the center of the earth is increased from 8500 feet to 9500 feet?

(A) 2944

(B) 8.127

(C) -123.047

(D) -0.00817

---

13. A membership to World Fitness costs \$75 per month and includes 10 free fitness classes. Any fitness classes attended after the first 10 free fitness classes cost \$5 each. Function C is used to model the cost of a monthly membership to World Fitness where n is the number of fitness classes taken and $C(n)$ is the cost in dollars. Which of the following defines C ?

(A) $C(n) = \begin{cases} 75, & 0 \le n \le 10 \\ 5n + 75, & n > 10 \end{cases}$

(B) $C(n) = \begin{cases} 75, & 0 \le n \le 10 \\ 5(n - 10) + 75, & n > 10 \end{cases}$

(C) $C(n) = \begin{cases} 75, & 0 \le n \le 10 \\ 10(n - 5) + 75, & n > 10 \end{cases}$

(D) $C(n) = 5n + 75$

FREE RESPONSE

Police use a formula to estimate the speed a car was traveling before an accident by measuring its skid marks. Function S is used to model the speed the car was traveling in mph where d is the distance the car skidded in feet and f is the coefficient of friction which depends on the road surface and road conditions.

$$S(d) = \sqrt{30df}$$

a. A country road has a coefficient of friction of 0.9 when it is dry and 0.4 when it is wet. What values of distance skidded would you expect in both conditions for a car that was travelling 110 mph when the brakes were applied?

b. Analyze the rate of change from a measured 70-foot skid mark to a 160-foot skid mark on the dry country road.

c. The distance of a skid mark is inversely proportional to the product of 30 and the coefficient of friction. Using the 60-foot skid mark on a dry country road, find the constant of proportionality. Explain what it means in this context.

1.14 Function Model Construction

AP Precalculus – Calculator Allowed Name: _____ **CA #1**

Graph the data and choose the regression that best fits the data.

1.

x	$f(x)$
14	48
18	50
28	53
33	64
35	70
38	75
41	84
48	105
53	138

a. Is the data Linear, Quadratic, or Cubic ?

b. Write the equation of the regression curve.

c. Use your equation to find the average rate of change from 20 to 40.

---

2. College students were asked how many classes they missed semester one and their grade in class.

Missed Classes	Grade %
1	88
1	91
3	82
4	79
6	73
6	71
8	65

a. Is the data Linear, Quadratic, or Cubic ?

b. Write the equation of the regression curve.

c. Use your equation to predict the grade of college student that misses 5 classes.

Construct an inversely proportional model to answer the following.

3. The value of y is inversely proportional to product of 4 and x squared. If $y = 12$ when $x = 2$, then find the value of y when $x = 6$

4. The time it takes to mow a lawn is inversely proportional to the number of workers. If it takes 3 workers 45 minutes to mow a lawn, how long will it take for 4 workers to mow the lawn?

Use the piecewise function to answer the following.

5.

$$f(x) = \begin{cases} \sqrt{x+1}, & 0 \le x \le 3 \\ 2x - 4, & 3 < x < 10 \\ 16, & 10 \le x \le 12 \end{cases}$$

a. Find $f(3)$.

b. Find $f(6)$.

---

6. A fast food company pays \$12 per hour for regular pay. Overtime pay is time and half.

$$P(h) = \begin{cases} 12h\,, & 0 \le h \le 40 \\ 18(h-40) + 480\,, & m > 40 \end{cases}$$

a. Find $P(48)$.

b. Find $P(40)$.

Answers to 1.14 CA #1

1. a. Quadratic b. $f(x) = 0.073x^2 - 2.754x + 73.845$ c. 1.626	2. a. Linear b. $f(x) = -3.487x + 92.873$ c. 75.438%
3. 1.333	4. 33.75 minutes
5. a. $f(3) = 2$ b. $f(6) = 8$	6. a. $P(48) = 624$ b. $P(40) = 480$

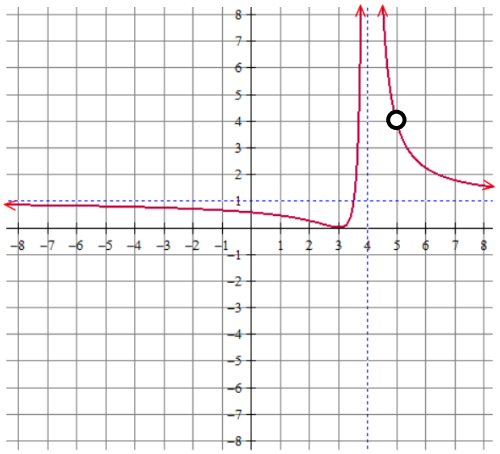
Name: _____ Date:_____ Period: _____ **1B Review**

Unit 1-B Review – Polynomial and Rational Functions

Reviews do NOT cover all material from the lessons but will hopefully remind you of key points. To be prepared, you must study all packets for lessons 1.7 – 1.14.

Use the graph of the rational function f to answer the following.

1.



a. Find the domain.

b. Find the zero(s).

c. Find the end behavior of f.

d. Find the limit notation of the vertical asymptote.

e. Find the limit notation of the hole.

Use the rational function to answer the following.

2.

$$g(x) = \frac{x^3 - 2x^2 - 15x}{x^2 + x - 6}$$

a. Find the domain.

b. Find the zero(s).

c. Find the y-intercept.

d. Find the hole(s) if there are any.

e. Find the vertical asymptote(s) if there are any.

f. Find the horizontal asymptote if there is one.

g. Find the slant asymptote if there is one.

Use the table of the rational function d to find the following.

3.

t	$d(t)$
-3.1	19
-3.01	199
-3.001	1,999
-3.0001	19,999
-3	undefined
-2.9999	$-21{,}111$
-2.999	-2.111
-2.99	-211
-2.9	-21

a. Find $\lim_{t\to -3^-} d(t) =$

b. Find $\lim_{t\to -3^+} d(t) =$

c. As t approaches negative three from the left the $d(t)\ldots$

d. As t approaches negative three from the right the $d(t)\ldots$

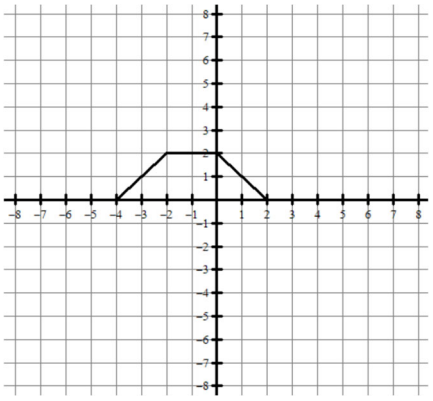
Use the binomial theorem to expand the following.

4. $(x+3)^5$

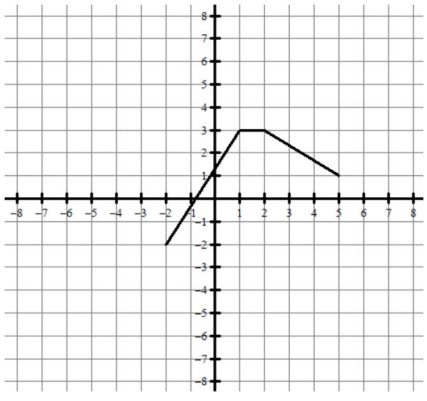
5. Find the 5th term in $(2x-1)^7$

GRAPHICAL TRANSFORMATIONS. Use the graph of f to graph $g(x)$.

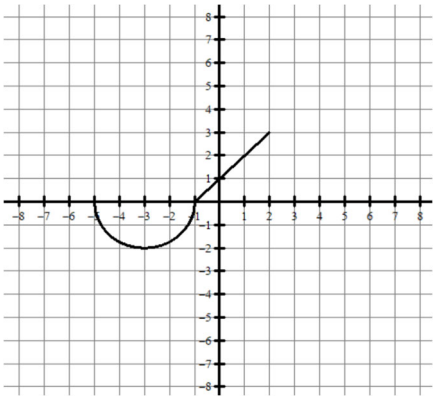
6. $g(x) = -f(x-2)$



7. $g(x) = 2f(x) - 4$



8. $g(x) = f(2x) + 5$



TRANSFORMATIONS. Use the give information to answer the following.

ALGEBRAICALLY

9. $f(x) = 6 - 4x^2$

$g(x) = f(x) + 5$, find $g(x)$.

NUMERICALLY

10.

Given the table of values for f.

x	$f(x)$
-6	2
-3	8
2	15
5	-2
8	-13

Let $g(x) = 3f(x) + 2$, find $g(5)$.

DOMAIN and RANGE

11.

Given the graph for f has a domain of $(-5,3)$ and range of $[-4, 8]$.

Let $g(x) = -f(x + 3) - 2$.

Find the domain and range of $g(x)$.

CALCULATOR ACTIVE. Graph the data and choose the regression that best fits the data.

12.

x	$f(x)$
14	48
18	50
28	53
33	64
35	70
38	75
41	84
48	105
53	138

a. Is the data Linear, Quadratic, or Cubic ?

b. Write the equation of the regression curve.

c. Use your equation to predict the value of $x = 34$.

CALCULATOR ACTIVE. Use the model to answer the questions in context.

13. A diver jumps from a cliff into the water below. The function $h(t) = -16t^2 + 14t + 80$ models the height of the diver over time, where *t* is time in seconds and *h* is the height in feet.

a. What is the restricted domain in this context?

b. What is the restricted range in this context?

Multiple Choice

14. The rational function f is given by $f(x) = \frac{2x^k(x+2)(x-5)}{3x^5-4x^2-2}$, where k is a positive integer. If the function f has a slant asymptote, what is the value of k ?

(A) 2

(B) 3

(C) 4

(D) 5

---

15. The rational function g is given by $g(x) = \frac{3x^5-2x^2+9x+5}{x(x-5)^2(x+7)}$. Which of the following describes the end behavior of g ?

(A) As x increases without bound, $g(x)$ increases without bound, and as x decreases without bound, $g(x)$ increases without bound.

(B) As x increases without bound, $g(x)$ increases without bound, and as x decreases without bound, $g(x)$ decreases without bound.

(C) As x increases without bound, $g(x)$ decreases without bound, and as x decreases without bound, $g(x)$ increases without bound.

(D) As x increases without bound, $g(x)$ decreases without bound, and as x decreases without bound, $g(x)$ decreases without bound.

---

16. The table gives values of $h(x)$ for selected values of x. Which of the following best describes $h(x)$?

x	$h(x)$
1.997	1.489
1.998	1.495
1.999	1.499
2	undefined
2.001	1.501
2.002	1.504
2.003	1.507

(A) The graph of h has a hole at $x = 2$

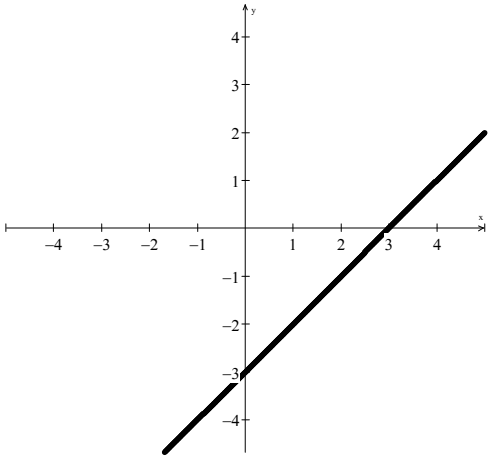
(B) The graph of h has a vertical asymptote at $x = 2$

(C) The graph of h has a hole at $x = \frac{3}{2}$

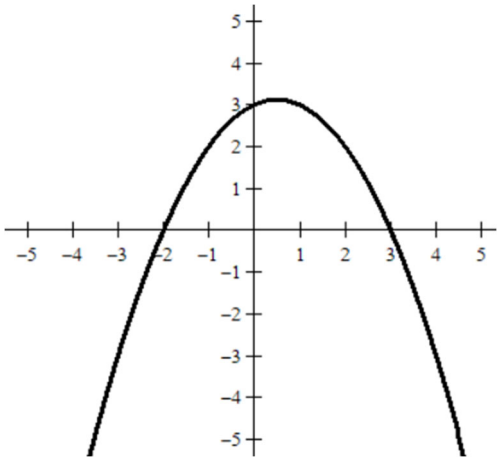
(D) The graph of h has a vertical asymptote at $x = \frac{3}{2}$

For questions 17 and 18, use the graphs of the polynomials f and g.

Graph of f



Graph of g



17. The function r is defined by $r(x) = \frac{f(x)}{g(x)}$. What is the value of the $r(0)$?

(A) -2

(B) 0

(C) -1

(D) 3

18. The function r is defined by $r(x) = \frac{f(x)}{g(x)}$. What are all the vertical asymptotes of the graph $y = r(x)$?

(A) at $x = -2$ only

(B) at $x = 3$ only

(C) at $x = -2$ and $x = 3$ only

(D) There are no vertical asymptotes.

Name: _____ Date:_____

Corrective Assignment

Unit 1B – Polynomial and Rational Functions

CALCULATOR ACTIVE for questions 1-4.

1. Use the function to answer the following.

$$f(x) = \frac{2x + 1}{x^2 - 3x - 10}$$

a. Domain:

b. Zero(s):

c. Hole(s):

d. Vertical Asymptote(s):

e. *y*-intercept:

f. Fill in the tables below.

x	$-10{,}000$	$-1{,}000$	-100	100	$1{,}000$	$10{,}000$
$f(x)$						

g. End Behavior:

h. Equation of the horizontal asymptote: (if one exists)

x	-2.1	-2.01	-2.001	-2	-1.999	-1.99	-1.9
$f(x)$							

i. $\lim_{x \to -2^-} f(x) =$

j. $\lim_{x \to -2^+} f(x) =$

2. Find the 3rd term when $(2x - 3)^6$ is expanded.

3. The value of a specific Beanie Baby over time is shown below starting with the year 1995.

Years since 1995	Value $
0	14
1	12
2	9
3	7.5
5	6
8	8
11	16
14	29
16	41
18	54
20	71

a. Is the data Linear, Quadratic, or Cubic ?

b. Write the equation of the regression curve.

c. Use your equation to predict the value of the Beanie Baby in 1999.

d. Find the average rate of change from 2000 to 2010.

4. F is inversely proportional to the square of r. If $F = 80$ when $r = 5$, find F when $r = 8$.

NO CALCULATOR for questions 5-25.

Vertical Asymptotes and Holes:

5. $f(x) = \frac{(x+4)(x-2)}{x^2-2x}$

a. Domain:

b. Hole(s):

c. Zero(s):

d. Vertical Asymptote(s):

6. $d(t) = \frac{t^2-9}{t+3}$

a. Domain:

b. Hole(s):

c. Zero(s):

d. Vertical Asymptote(s):

7. $r(x) = \frac{x^2-x-20}{2x-8}$

a. Domain:

b. Hole(s):

c. Zero(s):

d. Vertical Asymptote(s):

Horizontal and Slant Asymptotes:

8. $f(x) = \frac{4x^4-3x^3+4x+1}{5x^3-2x^2+1}$

Circle one:

The graph of f has a horizontal asymptote.

The graph of f has a slant asymptote.

The graph of f does not have a horziontal or slant asymptote.

9. $f(x) = \frac{2x^4+x^2+1}{3x^5-2x^2+5x}$

Circle one:

The graph of f has a horizontal asymptote.

The graph of f has a slant asymptote.

The graph of f does not have a horziontal or slant asymptote.

10. $f(x) = \frac{x^3+5x^2+x+2}{3x^3+2x^2-3}$

Circle one:

The graph of f has a horizontal asymptote.

The graph of f has a slant asymptote.

The graph of f does not have a horziontal or slant asymptote.

11. Find the equation of the slant asymptote.

$f(x) = \frac{4x^3-5x+3}{2x^2+3x}$

12. Use the function to answer the following.

$$f(x) = \frac{2x(x-5)}{x^2 - 2x - 15}$$

a. Domain:

b. Zero(s):

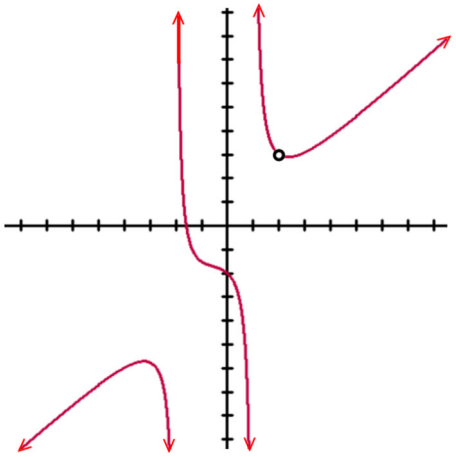
c. Hole(s):

d. Vertical Asymptote(s):

e. Horizontal Asymptote:

f. *y*-intercept:

13. Use the function to answer the following.



a. Domain:

b. Zero(s):
(Approximate)

c. *y*-intercept:

d. Limit Notation Hole(s):

e. End Behavior:

f. Limit Notation Vertical Asymptote(s):

g. Circle the statement that describes the graph:

The graph of f has a horizontal asymptote.

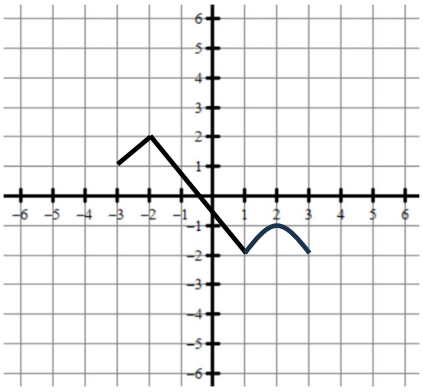
The graph of f has a slant asymptote.

The graph of f does not have a horziontal or slant asymptote.

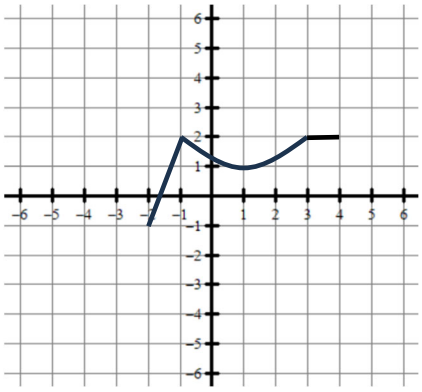
TRANSFORMATIONS

GRAPHICALLY

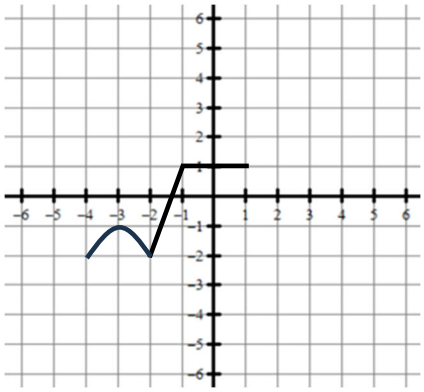
14. $g(x) = f(x-3) + 4$



15. $g(x) = -f(2x) + 5$



16. $g(x) = 3f(x-1)$



ALGEBRAICALLY

17. $f(x) = x^2 + 3x - 5$

$g(x) = f(x+2)$, find $g(x)$.

18. $f(x) = 2x^2 + 4x$

$g(x) = \frac{1}{2}f(x) - 5$, find $g(x)$.

19. $f(x) = 3x + 1$

$g(x) = -f(3x) + 4$, find $g(x)$.

NUMERICALLY

20. Given the table of values for f.

x	$f(x)$
-2	12
-1	18
0	5
1	-12
2	-3

Let $g(x) = f(-2x) + 2$,

find $g(1)$.

21. Given the table of values for f.

x	$f(x)$
0	12
1	9
2	6
3	3
4	0

Let $g(x) = -2f(x-1) - 3$,

find $g(1)$.

DOMAIN and RANGE

22.
Given the graph for f has a domain of $(-1,3)$ and range of $[-5,10]$.
Let $g(x) = 4f(x+1)$.
Find the domain and range of $g(x)$.

23.
Given the graph for f has a domain of $[-10,4]$ and range of $[-3,6]$.
Let $g(x) = f(2[x-2]) + 4$.
Find the domain and range of $g(x)$.

Answers to Unit 1B Corrective Assignment

1. a. $(-\infty,-2) \cup (-2,5) \cup (5,\infty)$
   b. $x = -\frac{1}{2}$
   c. none
   d. $x = -2$ and 5
   e. $y = -\frac{1}{10}$
   f. see below
   g. $\lim_{x\to-\infty} f(x) = 0$ $\lim_{x\to\infty} f(x) = 0$
   h. $y = 0$
   i. $\lim_{x\to-2^-} f(x) = -\infty$
   j. $\lim_{x\to-2^+} f(x) = \infty$

x	−10,000	−1,000	−100	100	1,000	10,000
$f(x)$	−0.0002	−0.002	−0.019	0.021	0.002	0.0002

x	−2.1	−2.01	−2.001	−2	−1.999	−1.99	−1.9
$f(x)$	−4.507	−43.081	−428.79	undefined	428.347	42.632	4.058

2. $2160x^4$

3. a. quadratic
   b. $f(x) = 0.298x^2 - 3.127x + 14.246$
   c. $f(4) = 6.505$
   d. 2.833 $ per year

4. $F = 31.25$

5. a. $(-\infty,0) \cup (0,2) \cup (2,\infty)$
   b. $x = 2$
   c. $x = -4$
   d. $x = 0$

6. a. $(-\infty,-3) \cup (-3,\infty)$
   b. $t = -3$
   c. $t = 3$
   d. No Vertical Asymptotes

7. a. $(-\infty,4) \cup (4,\infty)$
   b. No Holes
   c. $x = -4$ and 5
   d. $x = 4$

8. The graph of f has a slant asymptote.

9. The graph of f has a horizontal asymptote.

(answers continued on back)

Answers to Unit 1B Corrective Assignment

10. The graph of f has a horizontal asymptote.

11. $y = 2x - 3$

12.

a. $(-\infty, -3) \cup (-3,5) \cup (5, \infty)$

b. $x = 0$

c. $x = 5$

d. $x = -3$

e. $y = 2$

f. $(0, 0)$

13.

a. $(-\infty, -2) \cup (-2,1) \cup (1,2) \cup (2, \infty)$

b. $x \approx -1.6$

c. $(0, -2)$

d. $\lim_{x \to 2^-} f(x) = 3$ and $\lim_{x \to 2^+} f(x) = 3$

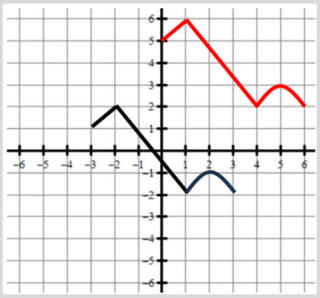
e. $\lim_{x \to -\infty} f(x) = -\infty$ and $\lim_{x \to \infty} f(x) = \infty$

f. $\lim_{x \to -2^-} f(x) = -\infty$ and $\lim_{x \to -2^+} f(x) = \infty$

$\lim_{x \to 1^-} f(x) = -\infty$ and $\lim_{x \to 1^+} f(x) = \infty$

g. The graph of f has a slant asymptote.

14.



15.



16.



17. $g(x) = x^2 + 7x + 5$

18. $g(x) = x^2 + 2x - 5$

19. $g(x) = -9x + 3$

20. $g(1) = 14$

21. $g(1) = -27$

22. Domain: $(-2,2)$
Range: $[-20,40]$

23. Domain: $[-3,4]$
Range: $[1,10]$

AP Precalc

Write your questions and thoughts here!

2.1 Change in Arithmetic and Geometric Sequences

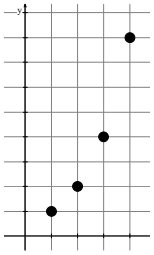
2.1 Notes

What is a sequence?

- A sequence is an ordered list of numbers. It could be finite or infinite.
- Each listed number is a term.

$\{1, 2, 4, 8, \ldots\}$

The graph of a sequence contains discrete points, not a connected line or curve.



Arithmetic Sequence

If each successive term in a sequence has a _____ (or a constant rate of change), the sequence is called **arithmetic**. The nth term of an arithmetic sequence is given by

$$a_n =$$

where ___ is the initial value [zero term] and ___ is the common difference.

1. Find an equation (rule) that represents the sequence $\{2, 5, 8, 11, \ldots\}$. Using your equation, find the 100th term of the sequence?

You can write the equation for the sequence using ANY term, not just the initial term.

$$a_n =$$

where _____ is the _____ term of the sequence.

2. What are other ways a rule can be written for the sequence $\{2, 5, 8, 11, \ldots\}$

***k*th term**	**Equation** $\boldsymbol{a_n = a_k + d(n - k)}$
$k = 1$	$a_n =$
$k = 2$	$a_n =$
$k = 3$	$a_n =$
$k = 4$	$a_n =$

3. Subtraction is still arithmetic. It is adding a negative number. Find an equation for the sequence $\{7, 2, -3, -8, \ldots\}$ using the 1st term $(k = 1)$ to set your rule.